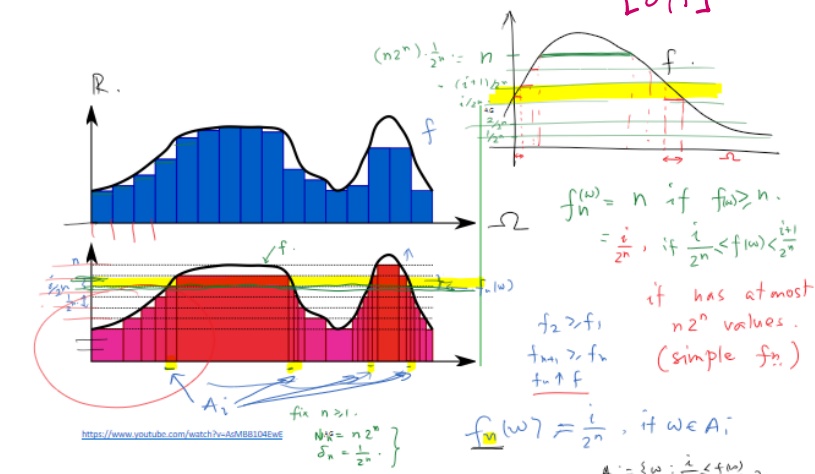
# Notes F

Monday, October 19, 2026 11:29 AM

$f = 1_{\mathbb{Q}\cap[0,1]}$, $U(f,P) = 1 \neq 0 = L(f,P)$ $\forall P$.

R-int doesn't exist. But $\int_{[0,1]} f\,dm = 1 \cdot m(\mathbb{Q}) = 0$.

$+0 \cdot m(\mathbb{Q}^c)$.




$f_n^{(\omega)} = n$ if $f(\omega) \geq n$.

$= \frac{i}{2^n}$, if $\frac{i}{2^n} \leq f(\omega) < \frac{i+1}{2^n}$

if has at most $n2^n$ values. (simple $f_n$)

$f_2 \geq f_1$

$f_{n+1} \geq f_n$

$f_n \uparrow f$

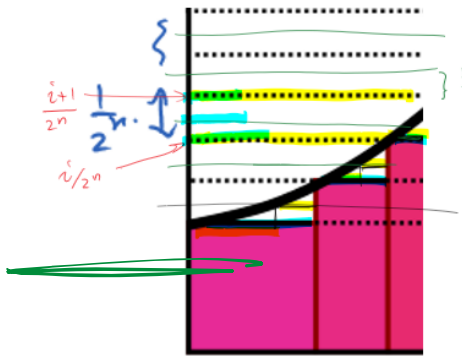
fix $n \geq 1$.

$\#\mathcal{P}_n = n2^n$

$\delta_n = \frac{1}{2^n}$.

$f_n(\omega) = \frac{i}{2^n}$, if $\omega \in A_i$

$A_i = \{\omega : \frac{i}{2^n} \leq f(\omega) < \frac{i+1}{2^n}\}$




$f_{n+1}(\omega) \geq f_n(\omega)$

if $\omega \in [\frac{i}{2^n}, \frac{i+1}{2^n}]$

$[\frac{2i}{2^{n+1}}, \frac{2(i+1)}{2^{n+1}}]$

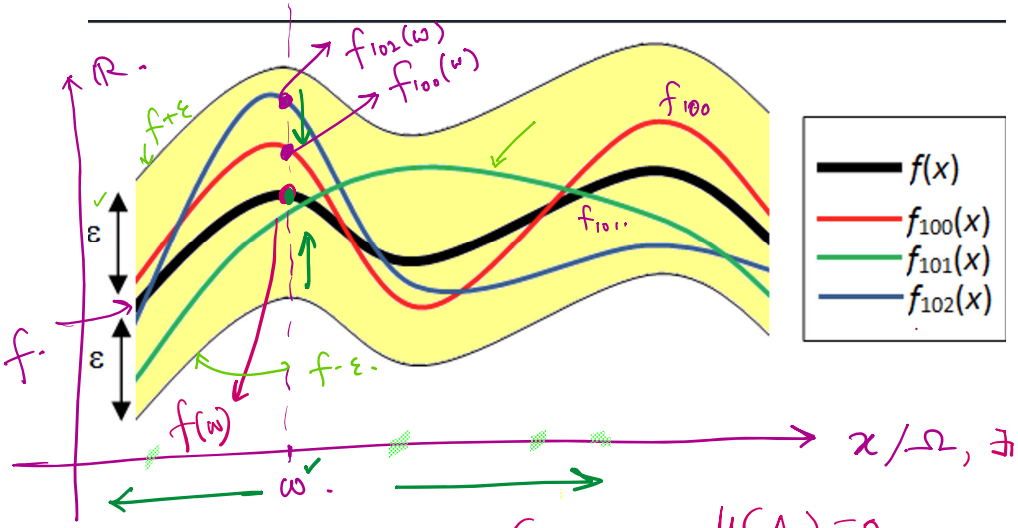
$[\frac{2i}{2^{n+1}}, \frac{2i+1}{2^{n+1}}] \cup [\frac{2i+1}{2^{n+1}}, \frac{2i+2}{2^{n+1}}]$

$\mathcal{C} = \{[a,b] \ldots\}$

# "Modes" of convergence (of functions).

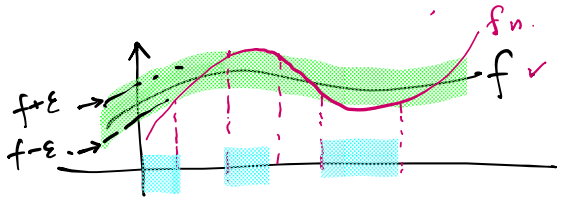
$a_n \to a \in \mathbb{R}$ as $n \to \infty$. $\Leftrightarrow$

$\forall \varepsilon > 0 \ \exists N$ s.t. $\forall n \geq N$, $|a_n - a| < \varepsilon$.




$x \in \Omega$, $\exists \mu$

$f_n \to f$ a.e. $(\mu)$ $\begin{cases} \mu(A) = 0 \\ \text{and } \forall \omega \notin A, \ f_n(\omega) \to f(\omega) \end{cases}$ on.




$\lim_{n \to \infty} \mu\left(\{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}\right) = 0$

$\forall \varepsilon > 0$.

$f_n \to f$ in measure $(\mu)$.

or $f_n \to f$ in probability if $\mu = P$.

conv

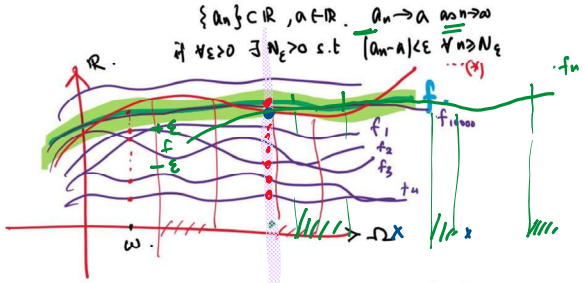
Friday, October 21, 2022 8:47 AM

$f_n, f$ real valued.

$\{a_n\} \subset \mathbb{R}$, $a \in \mathbb{R}$. $a_n \to a$ as $n \to \infty$

if $\forall \varepsilon > 0 \ \exists N_\varepsilon > 0$ s.t. $|a_n - a| < \varepsilon \ \forall n \geq N_\varepsilon$

$\Leftarrow$ conv. of reals (at the heart of it)




$\{f_n\}, f$ meas fns from $(\Omega, \mathcal{F}, \mu)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$f_n \to f$ at $\omega$ if $f_n(\omega) \to f(\omega)$ as $n \to \infty$ (as in (*))

$f_n \to f$ everywhere if true for all $x$. (pointwise)

$f_n \to f$ a.e. $(\mu)$ if $\forall\, \omega \notin A$, $(\mu(A) = 0)$

$f_n(\omega) \to f(\omega)$ as $n \to \infty$.

$\mu\left(\lim_{n\to\infty} f_n \neq f\right) = 0$ ; or, $P\left(\lim_{n\to\infty} X_n = X\right) = 1$

$f_n \xrightarrow{m} f$ if $\forall \varepsilon > 0$ $\mu(|f_n - f| > \varepsilon) \to 0$ as $n \to \infty$.

$\forall \varepsilon > 0$, i.e. $\forall \varepsilon' > 0 \ \exists N_{\varepsilon'} \ \forall \varepsilon' > 0$ s.t. $\mu(|f_n - f| > \varepsilon) < \varepsilon'$

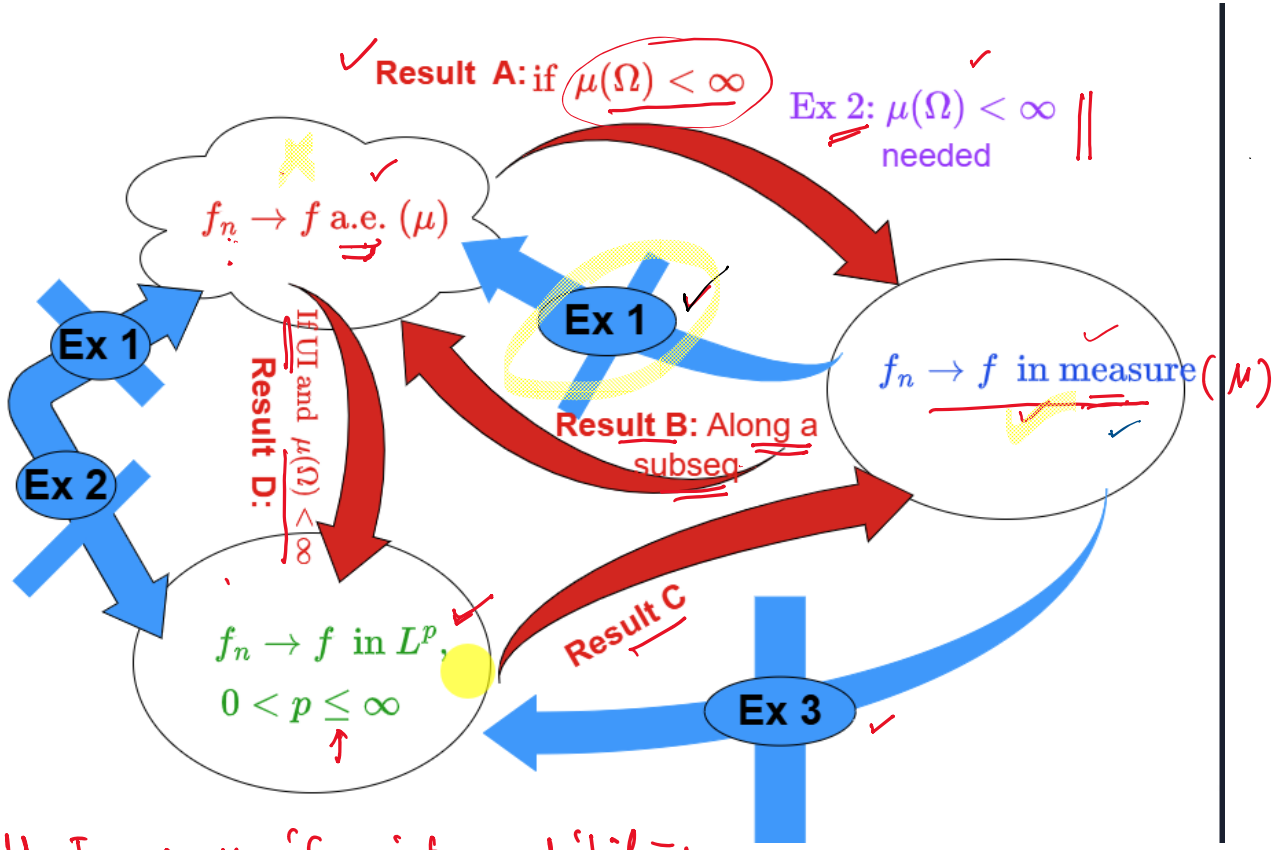
$\lim_{n\to\infty} \mu(|f_n - f| > \varepsilon) = 0 \ \forall \varepsilon > 0$

$\forall \varepsilon > 0 \ \exists N(\omega, \varepsilon) \geq 1$ s.t.

$|f_n(\omega) - f(\omega)| < \varepsilon$

$\forall n \geq N$.

a.e $\equiv$ a.s surely




U.I → unif. integrability.

$\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}([0,1])$, $\mu = m$

ex: (1)

typewriter sequence.




$f_1 = 1_{[0,1]}$ $\quad n = 1 = 2^0 + 0$ $\quad k = 0$ $\quad f \equiv 0$.

$f_2 = 1_{[0,\frac{1}{2}]}$, $f_3 = 1_{[\frac{1}{2},1]}$ $\quad k = 1$ ✓

$f_4 = 1_{[0,\frac{1}{4}]}$, $f_5 = 1_{[\frac{1}{4},\frac{1}{2}]}$, $f_6 = 1_{[\frac{1}{2},\frac{3}{4}]}$, $f_7 = 1_{[\frac{3}{4},1]}$ $\quad k = 2$

$5 = 2^2 + 1$

$k = 3$

fix $\omega \in \Omega = [0,1]$

$f_n(\omega) = 0$ or $1$ infinitely many times

$f \equiv 0$ ✓

$\mu(\{\omega : |f_n(\omega) - f(\omega)| > \varepsilon\}) = \frac{1}{2^k}$ ✓ $\quad 0 < \varepsilon < 1$

if $2^k \le n < 2^{k+1}$ $\quad$ read

$< \frac{2}{n} \to 0$ ✓ $\quad \frac{1}{2^k} = \frac{2}{2^{k+1}} < \frac{2}{n}$ ✓

as $n \to \infty$

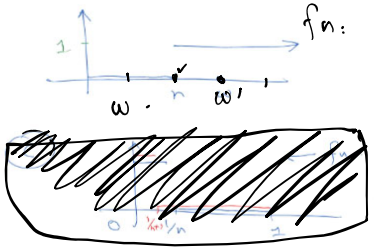
So $f_n \xrightarrow{\text{in meas}} f \equiv 0$ but $f_n \not\xrightarrow{a.e.} f$

ex (2)

$f_n = 1_{[n,\infty)}$, $(\Omega, \mathcal{F}, \mu) = (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+), m)$

(infinite indicators). $\quad [0,\infty)$

$f \equiv 0$ ✓




$f_n$: $\quad f \equiv 0$ ✓

$f_n \xrightarrow{a.e.(\lambda)} f$ but not in meas.

$f_n \xrightarrow{a.e.(\lambda)} f$ but not in $L_1$ ✓

* check the other statements in diagram.

take $\varepsilon > 0$. $\quad$ if $\varepsilon \ge 1$ then $\mu(|f_n - f| > \varepsilon) = 0$

if $0 < \varepsilon < 1$

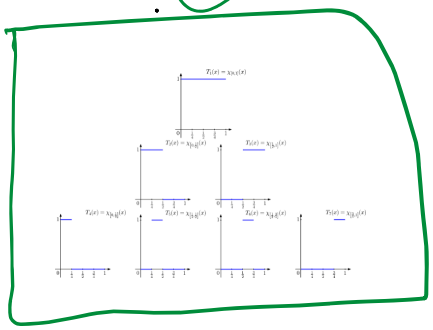
$\mu(|f_n - f| > \varepsilon) = \mu((n,\infty)) = \infty \not\to 0$ as $n \to \infty$.

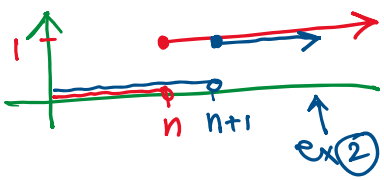
for $\omega \in \Omega = \mathbb{R}_+$

take $N > \omega$, $\quad f_n(\omega) = 0 \quad \forall n > N$.

$\to f(\omega) = 0$

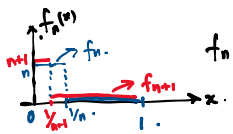
ex (1)





ex ③ — tall & thin indicator →

$(\Omega, \mathcal{F}, \mu) = ([0,1], \mathcal{B}([0,1]), m)$, $f \equiv 0$.




$f_n(x) = n \cdot 1_{[0,\frac{1}{n}]}(x)$.




"growing rectangle of area 1"

ex ③

$f_n(x) \equiv n \not\to f(x) \equiv 0$. when $x = 0$

but $\forall\, x \in (0,1]$, $f_n(x) = 0$ for

So, $f_n \to f$ a.e.$(\mu)$. since $\mu(\{0\}) = 0$

but $\int |f_n - f| d\mu = \int_0^1 n \cdot 1_{[0,\frac{1}{n}]} dm = n \cdot m([0,\frac{1}{n}])$

$= 1 \not\to 0$. So $f_n \not\to f$ in $L_1$.

**Thm 2.5.1:** $\mu(\Omega) < \infty$,

given $\mu\left(\lim_{n\to\infty} f_n \neq f\right) = 0$, show $\forall \varepsilon > 0$, $\lim_{n\to\infty} \mu(|f_n - f| > \varepsilon) = 0$

$\Updownarrow$

$\mu\left(\limsup_{n\to\infty} |f_n - f| > 0\right) = 0$ ($A$)

will show $\forall r \geq 1$, $\lim_{n\to\infty} \mu\left(|f_n - f| \geq \frac{1}{r}\right) = 0$

take any $\varepsilon > 0$, choose $r$

$\mu(|f_n - f| > \varepsilon) \leq \mu\left(|f_n - f| \geq \frac{1}{r}\right) \to 0$

$A = \{\omega : \limsup_{n\to\infty} |f_n(\omega) - f(\omega)| > 0\}$.

$= \{\omega : \limsup_{n\to\infty} |f_n(\omega) - f(\omega)| \geq \frac{1}{r} \text{ for some } r \geq 1\}$

$= \bigcup_{r\geq 1} \{\omega : \inf_{n\geq 1} \left(\sup_{j\geq n} |f_j(\omega) - f(\omega)|\right) \geq \frac{1}{r}\}$.

$= \bigcup_{r\geq 1} \{\omega : \sup_{j\geq n} |f_j(\omega) - f(\omega)| \geq \frac{1}{r}, \forall\, n \geq 1\}$

$= \bigcup_{r\geq 1} \bigcap_{n\geq 1} \{\omega : \sup_{j\geq n} |f_j(\omega) - f(\omega)| \geq \frac{1}{r}\}$

$= \bigcup_{r\geq 1} \bigcap_{n\geq 1} \bigcup_{j\geq n} \{\omega : |f_j(\omega) - f(\omega)| \geq \frac{1}{r}\}$.

$B_j(r) = \{\omega : |f_j(\omega) - f(\omega)| \geq \frac{1}{r}\}$

$C_n(r) = \bigcup_{j\geq n} B_j(r)$

$D(r) = \bigcap_{n\geq 1} C_n(r)$

$A = \bigcup_{r\geq 1} D(r)$ and $\mu(A) = 0$

$\mu(A) = 0 \Rightarrow \mu(D(r)) \leq \mu(A) = 0 \quad \forall\, r \geq 1$

$C_n(r) \downarrow D(r)$ as $n \to \infty$. and $\mu(\Omega) < \infty$

So mcfa applies : $\lim_{n\to\infty} \mu(C_n(r)) = \mu(D(r)) = 0$

Note $\mu(B_j(r)) \le \mu(C_n(r)) \quad \forall\, j \ge n$ .

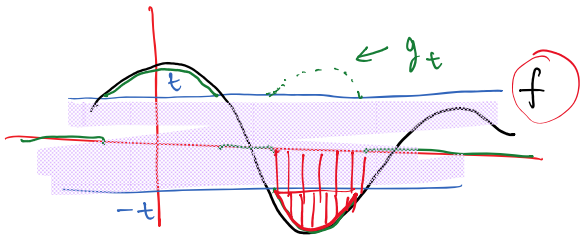
$\mu(B_n(r)) \le \mu(C_n(r)) \to 0$ as $n \to \infty$ $\forall\, r \ge 1$

$\lim_{n\to\infty} \mu(B_n(r)) = 0 \quad \forall\, r \ge 1$

i.e, $\lim_{n\to\infty} \mu\left(\left\{\omega : |f_n(\omega) - f(\omega)| \ge \frac{1}{r}\right\}\right) = 0$

pf complete.

## Concept of uniform integrability (u.i)




take $f \in L_1(\mu)$

consider

$$\int |f| \, 1_{\{|f|>t\}} \, d\mu \quad \checkmark$$

$g_t \to 0$ as $t \to \infty$

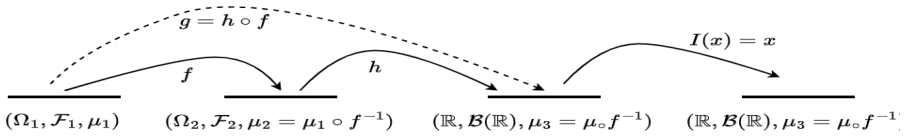
$f \in L_1 \Rightarrow |f| < \infty$ a.e $(\mu)$

$g_t = |f| \, 1_{\{|f|>t\}} \to 0$ a.e $(\mu)$ (use DCT Cont version)

$|g_t| \le |f|$, $f \in L_1$ so $\int g_t \, d\mu \to 0$ as $t \to \infty$.

take $f, g \in L_1$ $\{f, g\}$ unif. int. class.

$$\max\left( \int |f| \, 1_{\{|f|>t\}} \, d\mu, \int |g| \, 1_{\{|g|>t\}} \, d\mu \right) \to 0$$




$$\int_{\Omega_1} g d\mu_1 = \int_{\Omega_2} h d\mu_2 = \int_{\mathbb{R}} x d\mu_3.$$

$\mu f^{-1}(A) = \mu(f^{-1}(A))$

$\mu_1 \circ f = h \circ f$.

$i(x) = x$. (mble)

$\Omega_1, \mathcal{F}_1, \mu_1$ ① $\xrightarrow{f}$ $\Omega_2, \mathcal{F}_2, (\mu_1 \circ f^{-1}) = \mu_2$ ② $\xrightarrow{h}$ $\Omega_3 = \mathbb{R}, \mathcal{F}_3 = \mathcal{B}(\mathbb{R})$ ③ $\mu_3 = \mu_2 \circ h^{-1}$ $\xrightarrow{i}$ $\mathbb{R}, \mathcal{B}(\mathbb{R})$ ④

$$\int_{\Omega_1} i(h \circ f) d\mu_1 = \int_{\Omega_2} i(h(y)) d\mu_2 \circ f^{-1}(y) = \int_{\Omega_3 = \mathbb{R}} i(u) d\mu_0 (h \circ f)^{-1}(u) = \int_{\mathbb{R}} u \, d\mu_3(u)$$

$\mu_3 = \mu_0 (h \circ f)^{-1}$

**Claim:** for any $h, f$ mble.

$$\int_{\Omega_1} h(f) d\mu_1 = \int_{\Omega_2} h \, d(\mu_1 \circ f^{-1})$$

① ③ ② ③

$$\int 1_B d\mu = \mu(B)$$

proof: case I take if $h = 1_A$ (for some $A \in \mathcal{F}_2$)

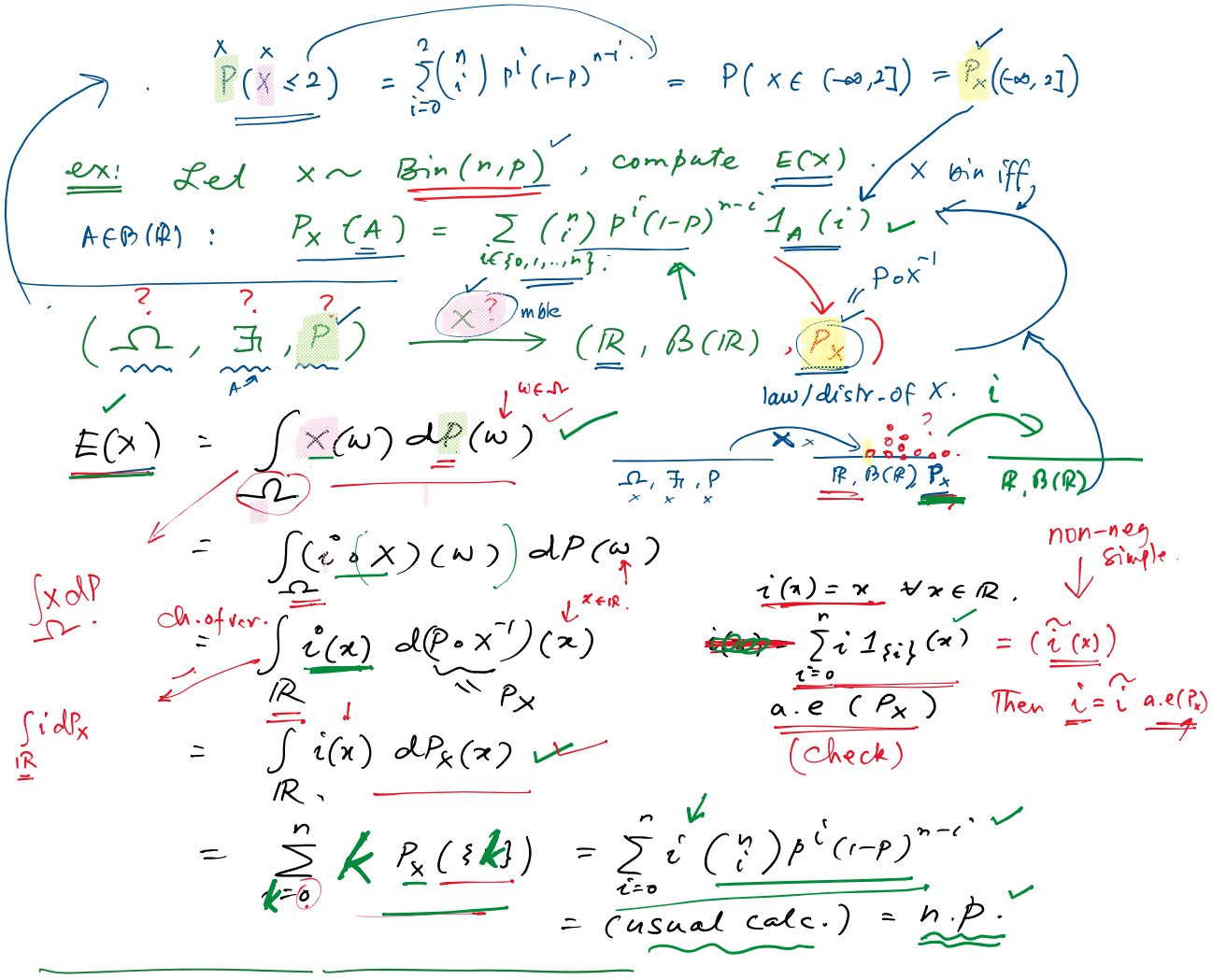
LHS $= \int_{\Omega_1} 1_A(f(\omega)) d\mu_1(\omega) = \int_{\Omega_1} 1_{f^{-1}(A)}(\omega) \, d\mu_1(\omega) = \mu_1(f^{-1}(A))$

RHS $= \int_{\Omega_2} 1_A \, d(\mu \circ f^{-1}) = (\mu \circ f^{-1})(A) = \mu(f^{-1}(A))$

So claim, holds for $h = 1_A$. ✓

can check that it holds for $h \geq 0$ simple } general approach for dealing with integral.

hence any $h \geq 0$, and then any (mble) $h$.


$P(X \le 2) = \sum_{i=0}^{2} \binom{n}{i} p^i (1-p)^{n-i}$
$= P(X \in (-\infty, 2]) = P_X((-\infty, 2])$
ex: Let $X \sim Bin(n,p)$, compute $E(X)$.
X bin iff,
$A \in \mathcal{B}(\mathbb{R})$ : $P_X(A) = \sum_{i \in \{0,1,\dots,n\}} \binom{n}{i} p^i (1-p)^{n-i} 1_A(i)$
$P \circ X^{-1}$
$(\Omega, \mathcal{F}, P)$ $\xrightarrow{X}$ $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$
mble
law/distr. of X.
$E(X) = \int_\Omega X(\omega)\, dP(\omega)$
$\omega \in \Omega$
$= \int_\Omega (i \circ X)(\omega)\, dP(\omega)$
$= \int_{\mathbb{R}} i(x)\, d(P \circ X^{-1})(x)$
$x \in \mathbb{R}$.
$= P_X$
$= \int_{\mathbb{R}} i(x)\, dP_X(x)$
$= \sum_{k=0}^{n} k\, P_X(\{k\}) = \sum_{i=0}^{n} i \binom{n}{i} p^i (1-p)^{n-i}$
$= (\text{usual calc.}) = n \cdot p$.
$\int X\, dP$
ch. of var.
$\int_{\mathbb{R}} i\, dP_X$
non-neg simple.
$i(x) = x \quad \forall x \in \mathbb{R}$.
$\sum_{i=0}^{n} i\, 1_{\{i\}}(x) = \tilde{i}(x)$
a.e $(P_X)$
(check)
Then $i = \tilde{i}$ a.e$(P_X)$

## Result (useful in Ch. 4).

In Cor. 2.3.6, we saw if $f \geq 0$ mble, then $(\Omega, \mathcal{F}, \mu)$.

$\nu(A) = \int_A f \, d\mu$ is a measure! $A \in \mathcal{F}$.

Take $h$ mble on $(\Omega, \mathcal{F})$. $\int_\Omega f \cdot 1_A \, d\mu$

then $\int h \, d\nu = \int h \cdot f \, d\mu$.

Pf: $h = 1_A \rightarrow \int h \, d\nu = \int 1_A \, d\nu = \nu(A)$ LHS.

$= \int_A f \, d\mu = \int 1_A \cdot f \, d\mu$ = RHS

LHS = $\int h \, d\nu = \int 1_A \, d\nu = 1 \cdot \nu(A) + 0 \cdot \nu(A^c) = \nu(A)$

$= \int h \cdot f \, d\mu$. proved for $h = 1_A$.

once it holds for $h = 1_A$, we can prove it for gen. meas. $h$ using usual steps.

---

ex: $X \sim N(0,1)$. find $E(X)$, $E(X^2)$.

$P_X(A) = \int_A \phi(x) \, dm(x)$ law of $X$.

$P_X((a,b]) = \Phi(b) - \Phi(a) = \int_a^b \phi(x) \, dx = \int_{(a,b]} \phi \, dm$

$E(X) = \int_\Omega X(\omega) \, dP(\omega) = \int_\Omega (i \circ X)(\omega) \, dP(\omega)$

$= \int_{\mathbb{R}} i(x) \, dP_X(x)$.

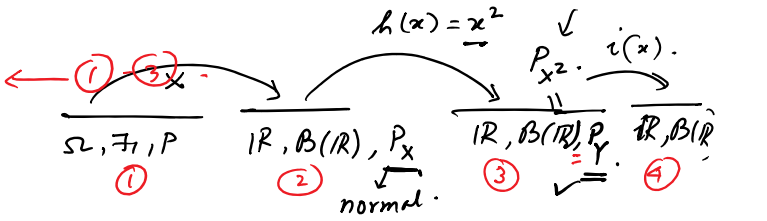
$= \int_{\mathbb{R}} x \, \phi(x) \, dm(x)$ Leb. int wrt leb. meas. (using prev. result.)

$= \int_{-\infty}^{\infty} x \, \phi(x) \, dx$ usual calc. (R-int.)

$= 0$.

normal calc. = 0

$E(X^2) =$




$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \quad -\infty < x < \infty.$$

$h(x) = x^2$

$= \int_\Omega h(X(\omega))\, dP(\omega)$

$= \int_{\mathbb{R}} h(x)\, dP_X(x)$ ② → ③

$= \int_{\mathbb{R}} x^2\, dP_X(x) = \int_{\mathbb{R}} x^2 \phi(x)\, dx$ ← R-int. (= leb. int.)

$= 1$.

$E(X^2) = \int h(X(\omega))\, dP(\omega)$ ③+④

$E(Y) = \int i(h(X(\omega)))\, dP(\omega)$

$Y = h(X) = X^2$.

$P_Y \to \chi^2_{(1)}$.

$E[\chi^2_{(n)}] = n$.

$Y = h(X) = X^2$

$= \int_{\mathbb{R}} i(y)\, dP_Y(y) = \int y \cdot g_{\chi^2_{(1)}}(y)\, dy = 1$.

Law of the unconscious statistician (LOTUS)

Real analysis fact:

$x_n \to x$ iff $\Longleftrightarrow$ $\forall$ subseq. $\{x_{n_k}\}$ of $\{x_n\}$, $\exists$ sub-subseq. $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ s.t $x_{n_{k_l}} \to x$. $\in \mathbb{R}$

proof. $\longrightarrow$

**DCT (modified)**

Let $f_n \xrightarrow{m} f$ and $|f_n| \le g$ a.e, $g \in L_1$!

then $\int |f_n - f| d\mu \to 0$

($x_n \to 0 = x$)

(also $\int f_n d\mu \to \int f d\mu$.)

Choose a subseq. $\{n_k\}$

$\int |f_{n_k} - f| d\mu$.

will show $\exists$ sub-subseq. $\{n_{k_l}\}$ s.t

$\int |f_{n_{k_l}} - f| d\mu \to 0$.

for this subseq. $f_{n_k} \xrightarrow{m} f$.

usual DCT

$\exists$ subseq s.t $f_{n_{k_l}} \xrightarrow{a.e} f$ (T.2.5.2) $\to$ U

(sub-subseq)

"Cont" version of DCT (MCT etc.)

Problem 2.22 read.

real analysis: $a_t \to a$ as $t \to \infty$

iff { $\forall \{t_n\}$, s.t. $t_n \to \infty$, we have $a_{t_n} \to a$. }

in DCT, $\{f_t\}_{t \in T=[0,\infty)}$ ex., $|f_t| \le g \ \forall t$, $g \in L_1$,

then $f_t \to f$ a.e. $\Rightarrow \int |f_t - f| d\mu \to 0$.

- need this for U.I.